

Attention U-Net for ocean oil-leakage detection

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Abstract

This report presents the work done during the Deep Learning 2026 project, on detecting ocean oil-leakages detection through binary segmentation done with Attention U-Net models.

Keywords: U-Net, Attention, Binary Segmentation, Oil, Ocean, Pollution.

1. Project objective

This project aims to develop and evaluate a high-precision deep learning-based semantic segmentation model for automated marine oil spill detection from Satellite Synthetic Aperture Radar (SAR) imagery, thereby supporting marine environmental monitoring.

Utilizing the Deep-SAR SOS dataset, this study addresses critical technical challenges, including inherent SAR speckle noise, sea-state interference, and severe class imbalance between oil spills and the vast ocean background.

The core objectives are as follows:

- **Efficient Data Pipeline:** Implement preprocessing to enhance training efficiency.
- **Optimized Feature Selection:** Integrate an Attention Gating (AG) mechanism into the classic UNET architecture to suppress redundant background noise from skip connections.
- **Class Imbalance Mitigation:** Employ a joint DiceBCELoss to maximize region overlap and prevent gradient vanishing caused by the overwhelming majority of background pixels.

Ultimately, this project delivers a computationally efficient and highly accurate deep learning pipeline for pixel-level oil spill detection. –

2. Dataset and image preprocessing

The dataset we chose is the Deep-SAR SOS Oil Spill Detection Dataset [1], combining satellite images of the Gulf of Mexico (ALOS PALSAR) as well as the Persian Gulf (Sentinel-1A). It is about 1.36Gb and contains 8,070 pairs of satellite imagery with their associated label, a binary segmentation of the image characterising the presence of oil in the sea.

SAR intensity images are converted to floating-point representations and normalised to the range $[0, 1]$ by scaling pixel values with a factor of $1/255$. All inputs are uniformly sized to a spatial resolution of 256×256 pixels – the theoretical original image size from the dataset – to ensure correct dimensions and adaptability of the model. The image resizing is done linearly to ensure fluid transitions and smoothening, while the mask resizing is done with a nearest neighbours interpolation to guarantee strictly binary boundaries of the label. The dataset is split with a standard 80/20 ratio between training and validation.

To optimize training and bypass disk I/O bottlenecks inherent to processing individual images at runtime, a preprocessing serialization strategy was adopted. The entire dataset is serialized beforehand into pre-batched PyTorch binary files (`.pt`). The `DatasetLoader` executes a RAM-mapping of these pre-computed tensor batches, eliminating the computational overhead of continuous runtime decoding and normalization.

So as to limit overfitting, we had to increase the size of the dataset through a process of data augmentation. While adding gaussian noise is a standard data augmentation practice in Deep Learning, it didn't seem relevant in this case due to the inherently noisy nature of the images caused by the satellite

imagery. Therefore, we augmented the dataset size by about 25% by adding rotations and flips of original satellite images. These transformations have been applied identically on the image and the label so as to preserve spatial coherence. The model’s generalisability to images from different satellite sensors is briefly discussed in the conclusion (Section 6).

3. Architecture

We selected the UNET architecture for this project. Indeed, UNET models are widely used in the literature for semantic segmentation tasks due to their efficiency in pixel-level labeling. Our network topology is inspired by the baseline architecture introduced by O. Ronneberger et al. [2] (see Fig. 1).

UNET is an encoder-decoder network where the encoder acts as a contracting path to extract features from the input image. It progressively increases the number of feature channels while reducing the spatial dimensions. To achieve this, it combines two Convolutional blocks—employing a 3x3 kernel and a ReLU activation function to handle non-linear relationships—with a max pooling operator for spatial downsampling.

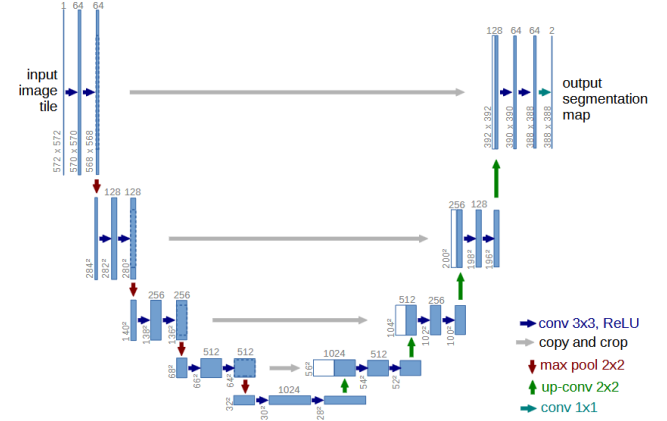


Figure 1: UNET architecture, from O. Ronneberger et al. [2].

Conversely, the decoder path designed by O. Ronneberger et al. consists of an upsampling step associated with a Convolutional block. The expansive path aims to combine the upsampled feature maps with the high-resolution feature channels extracted during the encoding phase. To do so, the spatial resolution is restored via transposed convolutions, followed by a skip connection to concatenate the decoder maps with the corresponding encoder features. Finally, a 1x1 convolution is applied to map the multi-channel feature maps into a single-channel output.

In our implementation, the architecture closely follows the parameters of Fig. 1 (in terms of network depth, kernel size, and activation functions), with the exception of the standard skip connections. The original mechanism transfers all extracted features indiscriminately, which lacks a clear feature hierarchy and can introduce spatial redundancy. To address this limitation, we integrated the attention gating mechanism introduced by O. Oktay et al. in [3] (see Fig. 2).

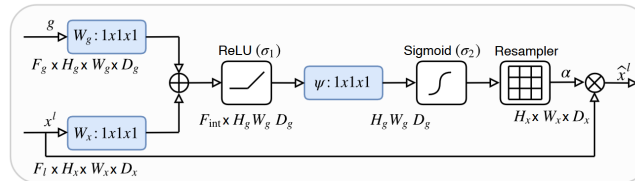


Figure 2: Attention Gate block diagram, adapted from O. Oktay et al. [3].

Assuming that both the gating vector g (from the decoder) and the skip feature map x (from the encoder) have already been aligned to share the exact same depth and spatial dimensions, the attention gating mechanism bypasses any preliminary resampling layer. The mathematical fusion and filtering process is defined by the following equation:

$$\alpha = \sigma(\psi(\text{ReLU}(x + g))) \quad (1)$$

Where:

- x represents the skip feature map from the encoder path.
- g represents the gating vector from the coarser decoder path.
- $+$ denotes the element-wise summation used to fuse the features of both vectors.
- ψ is a subsequent 1×1 convolution that compresses the fused multi-channel features into a single-channel space.
- σ represents the Sigmoid activation function.

The resulting matrix $\alpha \in [0, 1]$ contains the final attention coefficients. These coefficients act as a spatial weight matrix: values close to 1 highlight salient features and target regions, while values close to 0 actively suppress irrelevant background noise and artifacts before the features are concatenated with the decoder path.

4. Results

The Attention U-Net was trained for up to 30 epochs using the Adam optimizer, a `ReduceLROnPlateau` scheduler (patience = 5), and early stopping based on validation accuracy. `BCELoss` was used as the training objective, and per-epoch accuracy was tracked on both the training and validation sets.

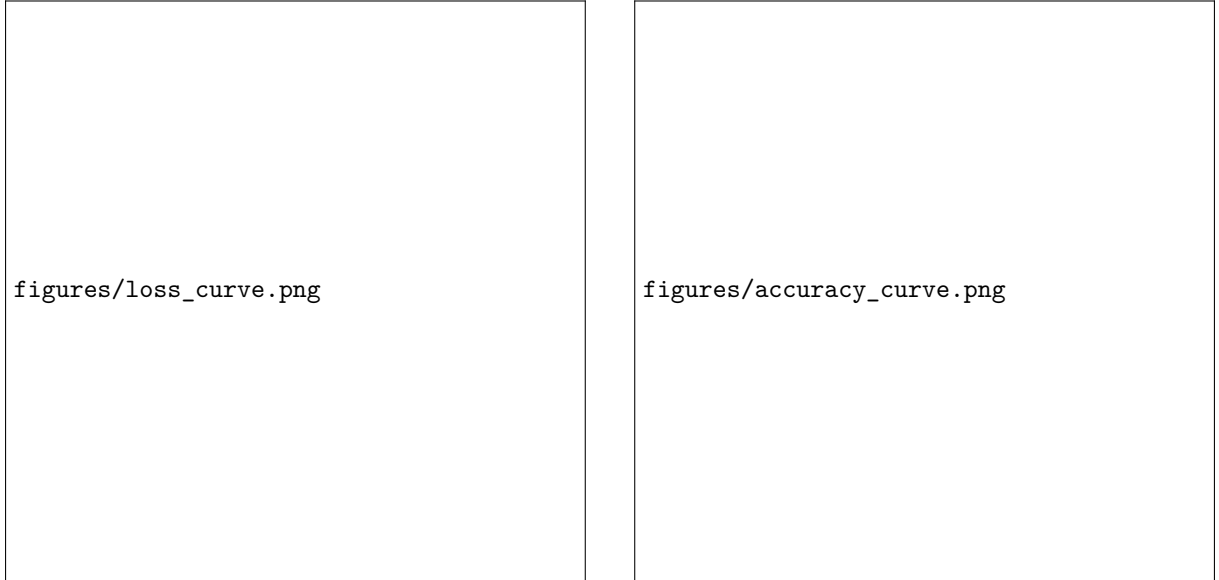


Figure 3: Training/validation loss (left) and accuracy (right) over epochs.

The final model reached a training loss of [X] and validation loss of [X], with a training accuracy of [X]% and validation accuracy of [X]%. Early stopping was triggered at epoch [XX].

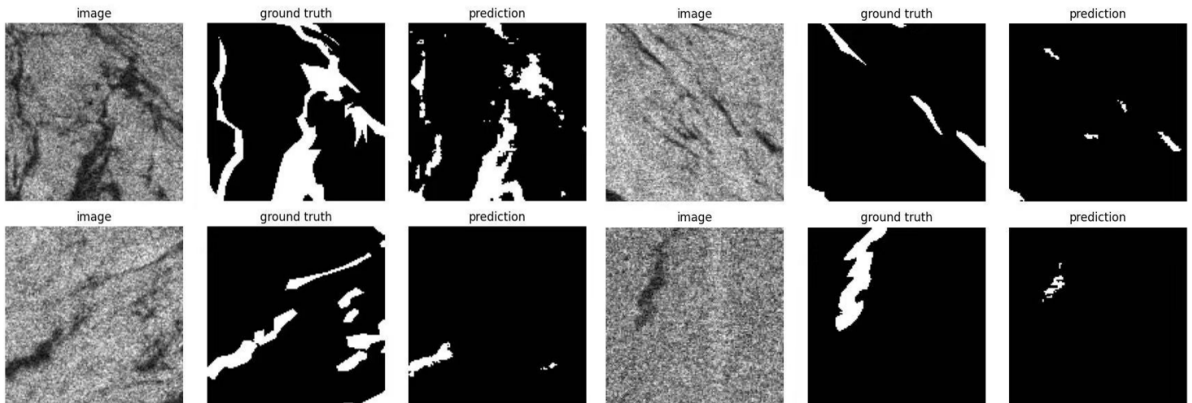


Figure 4: Predicted segmentation masks versus ground truth on validation samples.

5. Analysis

The loss and accuracy curves show [stable convergence / a growing train-validation gap / a plateau in later epochs], indicating that the model [generalizes well to unseen data / mildly overfits, contained by early stopping / is limited by capacity or data volume]. The learning-rate scheduler reduced the learning rate [X] time(s) during training, which [stabilized / had little visible effect on] validation performance afterward.

DiceBCELoss was chosen over plain cross-entropy because the SAR foreground/background pixels are imbalanced. The Dice term directly optimizes region overlap and helps mitigate the near-zero gradient problem often observed when using unweighted binary cross-entropy alone. This choice is reflected in the stability of the learning curves shown in Figure 3.

Qualitatively, the predictions in Figure 4 show that the model [captures large target regions accurately / struggles with thin or small structures / produces some false positives near boundaries], which is consistent with the quantitative trends described above.

6. Conclusion

The Attention U-Net, trained with BCELoss and validation-driven early stopping and learning-rate scheduling, achieved [summarize final performance] on the SAR binary segmentation task. The training pipeline, built on offline-preprocessed .pt data with paired training and validation data loaders, provides a stable and reproducible baseline. Future work could explore data augmentation to improve robustness on small or thin structures, as well as further tuning of the network depth and dropout rate to better balance the bias–variance trade-off observed in the learning curves.

Looking forward, incorporating adversarial domain adaptation techniques, as presented in [4], – particularly in the bottleneck – could successfully mitigate sensor-specific noise. This would grant our U-Net model the cross-satellite robustness required for operational oil spill detection, even when moving to new constellations without annotated data.

References

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Appendix